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Benign vs Malignant Ultrasound Image Classification

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1. Problem Statement and Dataset

1.1 Problem statement

The task is binary classification of ultrasound images into **benign** (label 0) and **malignant** (label 1, the positive class). Two models are compared on exactly the same held-out test images: a classical **baseline** (HOG features with a class-weighted RBF-kernel SVM) and an **improved** deep-learning model (ImageNet-pretrained MobileNetV2 with a new classification head, augmentation and class-weighted loss).

A single test set gives a single number per model. That number is a **sample statistic**, and a different random sample of images would give slightly different values. The statistical question is therefore whether the observed gap is larger than what sampling variability alone would produce.

Hypotheses (two-sided, significance level $\alpha = 0.05$):

- **H_0:** the two models have the same performance in the population of images the test set was drawn from (for paired accuracy: the probability that only the baseline is right equals the probability that only the improved model is right).
- **H_1:** the performance of the two models differs.

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- **H_1:** the performance of the two models differs.

Because malignant images are the majority class, accuracy alone can be misleading. The analysis therefore also reports balanced accuracy, malignant recall (sensitivity), specificity, F1, ROC-AUC and PR-AUC.

1.2 Dataset

The data come from a ZIP archive (Datasets.zip) with folders **benign/**, **malignant/** and **normal/**. The "normal" folder is ignored, so only benign and malignant images are used. Duplicate file paths are removed and only image files (.png, .jpg, .jpeg, .bmp, .tif, .tiff) are kept.

Subset	Images (n)	Malignant	Benign	% malignant
Whole dataset	2,599	2,209	390	85.0%
Train	1,558	1,325	233	85.0%
Validation	524	445	79	84.9%
Test (held out)	517	439	78	84.9%

Table 1. Dataset and split sizes. Class proportions are almost identical across subsets (stratification worked).

Class imbalance. About 85% of images are malignant. A trivial rule that always outputs "malignant" would be correct on 439 of 517 test images, an accuracy of **0.849**. Every accuracy figure in this report must be read against that reference level.

Data-quality observations (important for interpretation). The sample grid displayed in the notebook raises two concerns that should be checked before any result is trusted:

- Three of the eight displayed images are **mask or outline images** (two thin ellipse outlines and one white blob on a black background), all labelled malignant. The executed notebook does not filter files whose names contain "mask" or "annotation".
- Several displayed images show an oval, skull-like structure that resembles a fetal-head ultrasound rather than a lesion. The provenance of the benign/malignant labels should be confirmed.

2. Sampling and Experimental Methodology

2.1 Population, sample and estimators

The 2,599 labelled images are treated as a sample from a larger population of similar ultrasound images. Each metric computed on the 517-image test set (accuracy, recall, AUC and so on) is an **estimator** of the corresponding true population value. Its uncertainty is described by its **sampling distribution**, which we approximate analytically (binomial theory) and empirically (bootstrap).

2.2 Sampling design: stratified, group-aware, one-time test set

- **Stratification.** StratifiedGroupKFold keeps the benign:malignant ratio similar in every subset (Table 1).
- **Grouping to limit leakage.** Images sharing the filename prefix before the first underscore are assumed to come from the same source and are kept in the same subset. The code verifies zero group overlap between train/test and validation/test (both 0).
- **Nested splitting.** A 5-fold split provides the test fold (the fold whose class mix is closest to the overall mix). The remaining data are split by a 4-fold group-aware split into train and validation.
- **Same test images for both models.** Predictions for both models are produced on the identical 517 images in the same order, which makes a **paired** comparison possible and valid.
- **Test set used once.** Training, early stopping and model selection use only train and validation data. The fixed seed (42) and the saved split CSV files make the split reproducible.

2.3 Models compared

	Baseline	Improved
Input	128×128 grayscale	224×224 RGB, MobileNetV2 preprocessing

	Baseline	Improved
Features / backbone	HOG (9 orientations, 8×8 cells, 2×2 blocks), standardised	ImageNet-pretrained MobileNetV2, global average pooling, dropout 0.35
Classifier	RBF SVM, C = 3, class_weight = balanced, Platt-scaled probabilities	Dense(1) + sigmoid
Imbalance handling	Balanced class weights	Class-weighted loss (benign ≈ 3.34, malignant ≈ 0.59)
Training	Fit on train set	Adam (lr 1e-3), augmentation (flip, rotation, zoom, contrast), early stopping on validation PR-AUC, LR reduction, then a second fine-tuning stage with early stopping (best weights restored)
Decision threshold	0.5	0.5 (not tuned on validation data)

Table 2. Model configurations as run in the executed notebook.

2.4 Metrics

With TP, TN, FP, FN taken from the confusion matrix (positive = malignant):

- **Accuracy** = (TP + TN) / n. **Sensitivity (recall)** = TP / (TP + FN). **Specificity** = TN / (TN + FP).
- **Precision** = TP / (TP + FP). **F1** = 2·Precision·Recall / (Precision + Recall). **Balanced accuracy** = (Sensitivity + Specificity) / 2.
- **ROC-AUC** = probability that a randomly chosen malignant image receives a higher score than a randomly chosen benign image (0.5 = chance). **PR-AUC** (average precision) summarises precision against recall; its "no-skill" level equals the malignant prevalence, 0.849 here.

2.5 Statistical methods used

Method	What it does	Applied to
Binomial model and standard error	Number of correct predictions ~ Binomial(n, p); $SE = \sqrt{p(1-p)/n}$	Accuracy, sensitivity, specificity
Confidence intervals	Wilson / Clopper–Pearson (exact binomial) intervals for proportions; percentile bootstrap (2,000 resamples of test images) for metrics without a simple formula	All metrics (Table 4)
McNemar test (paired)	$\chi^2 = (\lvert b - c \rvert - 1)^2 / (b + c)$, df = 1, where b = only baseline correct and c = only improved correct; exact binomial version as a check	Per-image correctness, same test images
Two-proportion test (Fisher exact)	Compares two proportions (conservative here, because it ignores pairing)	Sensitivity, specificity
AUC comparison (z-test)	Hanley–McNeil standard errors; $z = (AUC_2 - AUC_1) / SE(diff)$, with $SE(diff)^2 = SE_1^2 + SE_2^2 - 2r \cdot SE_1 \cdot SE_2$ where r is the correlation between the two AUC estimates	ROC-AUC
Multiple-comparison check	Bonferroni threshold $\alpha / 4 = 0.0125$ for the four headline tests	Interpretation of p-values

Table 3. Statistical toolkit.

3. Observations

3.1 Test-set performance (n = 517; 439 malignant, 78 benign)

Metric	Baseline: HOG + SVM	Improved: MobileNetV2	Difference (Imp. − Base)
Accuracy	0.8491	0.6228	−0.2263
Balanced accuracy	0.5422	0.7726	+0.2304
Precision (malignant)	0.8603	0.9959	+0.1356
Recall / sensitivity (malignant)	0.9818	0.5581	−0.4237
Specificity (benign)	0.1026	0.9872	+0.8846
F1 (malignant)	0.9170	0.7153	−0.2017
ROC-AUC	0.7660	0.8261	+0.0601
PR-AUC	0.9541	0.9670	+0.0129
TN / FP / FN / TP	8 / 70 / 8 / 431	77 / 1 / 194 / 245	

Table 4. Point estimates at threshold 0.5 (values from the notebook output; differences computed here).

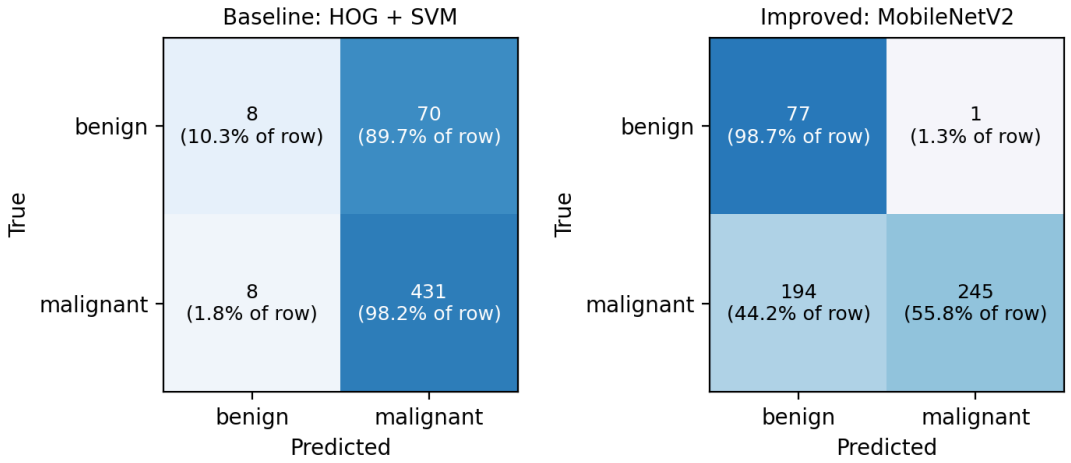


Figure 1. Confusion matrices on the same 517 test images (percentages are of each true class).

What the confusion matrices show.

- The **baseline** predicts “malignant” for 501 of 517 images (96.9%). It finds 431 of 439 malignant cases but recognises only 8 of 78 benign images. Its 439 correct answers (8 + 431) equal the 439 you would get by always answering malignant, so its accuracy contains **no information beyond the class prior**. Its precision of 0.860 is barely above the prevalence of 0.849 (a lift of 1.01).
- The **improved model** predicts malignant for only 246 images (47.6%). It is almost never wrong when it says malignant (1 false positive), and it recognises 77 of 78 benign images, but it **misses 194 of 439 malignant cases (44.2%)**. This pattern is typical of a class-weighted model whose probability scale has shifted toward “benign” and whose threshold was left at 0.5.

3.2 Estimation: how precise are these numbers?

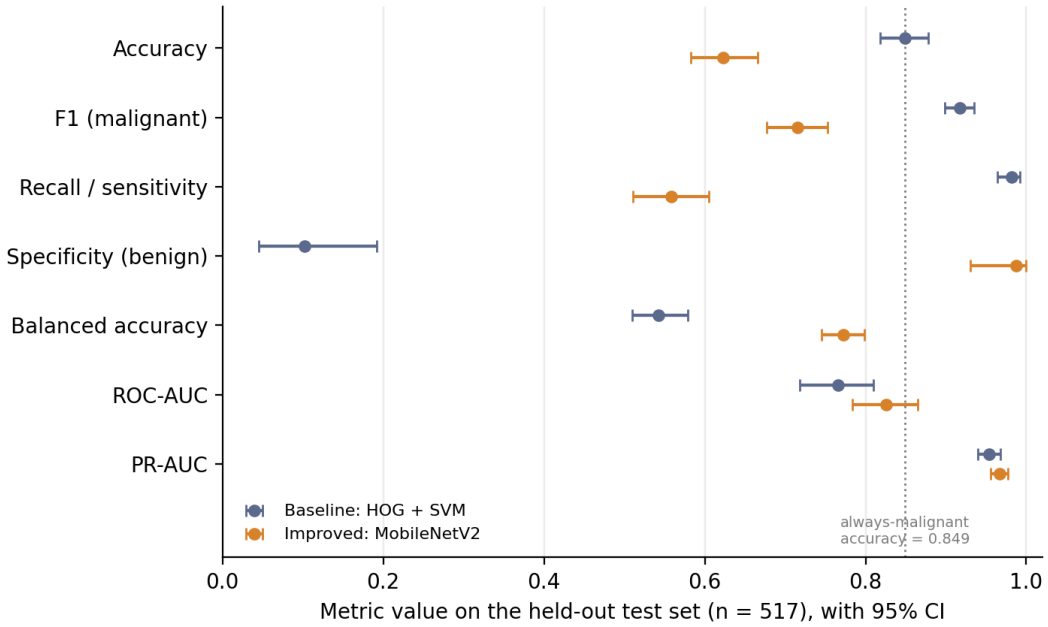


Figure 2. Point estimates with 95% confidence intervals. The dotted line marks the accuracy of always predicting malignant.

Metric	Baseline	95% CI	Improved	95% CI	CI method
Accuracy	0.849	0.818 – 0.878	0.623	0.582 – 0.665	Bootstrap
F1 (malignant)	0.917	0.899 – 0.935	0.715	0.678 – 0.753	Bootstrap
Recall (sensitivity)	0.982	0.964 – 0.992	0.558	0.510 – 0.605	Clopper–Pearson
Specificity	0.103	0.045 – 0.192	0.987	0.931 – 1.000	Clopper–Pearson
Balanced accuracy	0.542	0.510 – 0.579	0.773	0.745 – 0.798	Binomial simulation
ROC-AUC	0.766	0.719 – 0.810	0.826	0.784 – 0.865	Bootstrap
PR-AUC	0.954	0.940 – 0.967	0.967	0.956 – 0.977	Bootstrap

Table 5. Bootstrap intervals are copied from the notebook. Clopper–Pearson and balanced-accuracy intervals were computed here from the confusion-matrix counts (20,000 simulated draws for balanced accuracy).

Worked example (normal approximation). For the improved model, $\hat{p} = 322/517 = 0.6228$, so SE = $\sqrt{(0.6228 \times 0.3772 / 517)} = 0.0213$ and the 95% interval is $0.6228 \pm 1.96 \times 0.0213 = [0.581, 0.665]$, matching the bootstrap interval. For the baseline, $\hat{p} = 0.8491$ and SE = 0.0157, giving [0.818, 0.880].

Reading the intervals. The accuracy, F1, recall, specificity and balanced-accuracy intervals of the two models are far apart and do not overlap, so these differences are not plausibly due to sampling. The ROC-AUC intervals overlap (0.719–0.810 vs 0.784–0.865) and the PR-AUC intervals overlap almost completely. Overlap does not by itself mean “no difference”, because the intervals ignore that both models were scored on the same images, so a paired test is needed.

The benign class has only 78 test images, so specificity is estimated much less precisely than sensitivity. The improved model’s specificity of 77/78 is consistent with a true value anywhere from about 0.93 to 1.00.

3.3 Hypothesis tests on the same test cases

McNemar test (accuracy, paired). The notebook's McNemar output was cut off in the exported PDF, so the exact paired counts are not visible. They can still be bounded rigorously from the two confusion matrices. The baseline is correct on 439 images and the improved model on 322, so **$b - c = 439 - 322 = 117$ exactly** (b = only baseline correct, c = only improved correct). Checking the benign and malignant classes separately gives $186 \le b \le 195$ and $69 \le c \le 78$, hence $b + c$ lies between 255 and 273.

Across that entire range the continuity-corrected statistic is $\chi^2 = (117 - 1)^2 / (b + c)$, which is between 49.3 and 52.8 on 1 degree of freedom, and the p-value is between 4×10^{-13} and 2×10^{-12} (the exact binomial version gives $p < 10^{-12}$). **The conclusion therefore does not depend on the missing counts:** the baseline is significantly more accurate than the improved model at threshold 0.5.

Comparison	Test	Statistic	p-value
Accuracy (paired)	McNemar, df = 1	$\chi^2 \approx 49 – 53$	$< 10^{-11}$
Sensitivity: 431/439 vs 245/439	Fisher exact (unpaired)	odds ratio ≈ 43	$\approx 10^{-58}$
Specificity: 8/78 vs 77/78	Fisher exact (unpaired)	odds ratio ≈ 0.0015	$\approx 10^{-33}$
ROC-AUC, r = 0 (conservative)	z-test, Hanley–McNeil SE	z = 1.87, SE = 0.032	0.062
ROC-AUC, r = 0.3	z-test	z = 2.22, SE = 0.027	0.026
ROC-AUC, r = 0.5	z-test	z = 2.62, SE = 0.023	0.009
ROC-AUC, r = 0.7	z-test	z = 3.34, SE = 0.018	0.0008
PR-AUC	CI overlap (diff = 0.013)	not significant	not tested formally

Table 6. Hypothesis tests. Fisher tests treat the two models' predictions as independent samples, which is conservative for paired data. ROC-AUC differences depend on r, the correlation between the two models' AUC estimates, because the same images are scored.

ROC-AUC difference. The observed difference is +0.060. Hanley–McNeil standard errors are 0.025 (baseline) and 0.021 (improved). If the two AUC estimates were independent ($r = 0$), the difference would have $p = 0.062$ and a 95% CI of $[-0.003, 0.123]$. Because both models were scored on the same images, a positive correlation is expected, and for r between 0.3 and 0.7 the difference is significant (p between 0.026 and 0.0008). This is why the result is called **suggestive but not conclusive**: the exact value of r is not known from the exported outputs. A paired bootstrap or DeLong test on the saved test probabilities settles it (code in Appendix A).

Multiple comparisons. With four headline tests the Bonferroni threshold is 0.0125. The accuracy, sensitivity and specificity results pass it by many orders of magnitude. The ROC-AUC result passes only if $r \gtrsim 0.4$.

3.4 Consistency across independent samples and training behaviour

- **Replication on validation data.** The baseline's validation report (n = 524) already showed benign recall 0.08, malignant recall 0.99 and accuracy 0.85, essentially the same pattern as on the test set (0.10, 0.98, 0.85). An effect that appears in two independent samples is unlikely to be a sampling fluke.
- **Little evidence of real learning gains during fine-tuning.** Validation PR-AUC stays essentially flat (about 0.965 to 0.968) across epochs while training PR-AUC climbs to about 0.974, and validation loss fluctuates between roughly 0.47 and 0.59. In the second stage the best epoch

was the first one, so early stopping restored those weights. This is a typical sign that extra training mainly fits the training set.

- **Context for PR-AUC.** Because malignant prevalence is 0.849, PR-AUC cannot fall below about 0.85 for any reasonable model. Values of 0.954 and 0.967 therefore indicate much smaller real differences than the raw numbers suggest.

4. Decision

Claim	Verdict	Basis
"Improved model has higher accuracy"	Rejected: the opposite is true	Baseline 0.849 vs 0.623; McNemar $p < 10^{-11}$; non-overlapping CIs
"Improved model finds malignant cases better"	Rejected	Recall 0.558 vs 0.982; 194 missed malignant cases vs 8; Fisher $p \approx 10^{-58}$
"Improved model recognises benign cases better"	Supported (large, significant effect)	Specificity 0.987 vs 0.103; Fisher $p \approx 10^{-33}$; wide CI for baseline (0.045–0.192)
"Improved model has better balanced accuracy"	Supported	0.773 (0.745–0.798) vs 0.542 (0.510–0.579); intervals do not overlap
"Improved model ranks cases better (ROC-AUC)"	Suggestive, not conclusive	+0.060; p = 0.062 (r = 0) to 0.0008 (r = 0.7); needs paired test
"Improved model has better PR-AUC"	Not supported	+0.013; CIs nearly identical; low ceiling because prevalence = 0.849

Table 7. Verdict for each claim.

Decision

The observed differences are **real, not due to test-sample variation**, but they reflect different **operating points** rather than a clear improvement. The baseline behaves almost like "always malignant". The improved model behaves like a conservative classifier that labels many malignant cases benign.

We therefore **do not conclude that the improved model is better**. The only defensible statement is that it shows modestly better threshold-independent discrimination (ROC-AUC) and balanced accuracy, with the AUC gain still needing a paired test for confirmation.

5. Limitations

- **Scope of the evidence.** This report analyses the executed notebook shown in mod.pdf (HOG + SVM vs MobileNetV2). The separately uploaded .ipynb is a different, un-run version (HOG + logistic regression vs EfficientNet-B0) and was not used for any numbers. The PDF export also omits one page, so the heading and details of the second training stage, and the exact McNemar and later output lines, are not visible. Where this mattered, results were derived from the confusion matrices.
- **One split, one seed.** Bootstrap intervals capture variation due to which test images were drawn, but not variation due to training randomness, initialisation or the choice of split. Results could shift with another seed or fold.
- **Small minority class.** Only 78 benign test images exist, so specificity and any benign-related metric have wide uncertainty, and the bootstrap percentile method is less reliable with so few minority samples.

- **Threshold not tuned.** Both models were evaluated at 0.5. Accuracy, F1, precision and recall at a fixed threshold mix ranking quality with the operating point, and class-weighted training shifts probability scales. The 0.5 results should not be read as the best either model can do.
- **Possible leakage and data-quality problems.** Grouping by filename prefix is an unverified proxy for patient identity. Mask and outline images labelled as malignant, and images resembling fetal-head scans, suggest dataset issues that could bias both models and inflate or distort any conclusion. Results are only as reliable as the labels.
- **Approximate AUC test.** The AUC z-test uses unpaired Hanley–McNeil standard errors adjusted by an assumed correlation r. It is indicative, not exact. Fisher tests for sensitivity and specificity ignore pairing and are conservative.
- **Multiple testing and no external validation.** Many metrics were examined, and no independent hospital, scanner or time-period test set was used. Performance on new data may be lower.
- **Not a clinical tool.** Nothing here supports clinical use.

6. Where Probability and Statistics Are Used

Concept	Where it appears in this project	Why it matters
Population vs sample	2,599 images treated as a sample; 517-image test set as a sample from the population	Test metrics are estimates, not exact truths
Stratified (group) random sampling	StratifiedGroupKFold with seed 42; class proportions ≈ 85/15 in every subset	Representative, reproducible, leak-resistant splits
Pseudo-randomness	Seeds for shuffling, augmentation, dropout, bootstrap	Reproducibility of every random step
Probability as model output	SVM Platt-scaled probability and sigmoid output = P(malignant | image); threshold 0.5 as a decision rule	Turns scores into decisions
Conditional probability and Bayes' rule	Sensitivity = P(pred + | malignant); specificity = P(pred − | benign); precision = P(malignant | pred +)	Precision depends on prevalence: baseline precision 0.860 vs prevalence 0.849
Binomial distribution and CLT	Correct predictions ~ Binomial(n, p); SE = $\sqrt{(p(1-p)/n)}$; normal-approximation CI	Quantifies uncertainty of accuracy, recall, specificity
Confidence intervals	Wilson, Clopper–Pearson, percentile bootstrap	Show the plausible range of each metric
Resampling (bootstrap)	2,000 resamples of test images with replacement	Approximates the sampling distribution of F1 and AUC
Hypothesis testing	McNemar, Fisher exact, z-test for AUC; H_0/H_1, α = 0.05, p-values	Decide whether differences exceed chance
Type I error and multiple comparisons	Bonferroni threshold 0.0125 for four tests	Controls false "significant" findings
Effect size	Differences in accuracy (−0.226), AUC (+0.060), PR-AUC (+0.013); odds ratios	Separates statistical from practical significance
Probabilistic meaning of AUC	AUC = P(score of a random malignant > score of a random benign); equivalent to the Mann–Whitney U statistic	Threshold-free comparison of models
Maximum likelihood / loss	Binary cross-entropy is the negative log-likelihood of a Bernoulli model; class weights re-weight the likelihood	Basis of training the CNN and handling imbalance
Random regularisation	Dropout (Bernoulli masks), random flips, rotations, zoom, contrast	Reduces overfitting through injected randomness

Appendix A. Code to finish the paired analysis in Colab

Run these cells after the evaluation section of the notebook (they reuse y_test, baseline_test_prob and improved_test_prob). They produce the exact McNemar counts, a paired bootstrap confidence interval for the difference in ROC-AUC and PR-AUC, and a threshold chosen on validation data only.

A.1 Exact paired counts and bootstrap of metric differences

```
import numpy as np
from sklearn.metrics import roc_auc_score, average_precision_score
from scipy.stats import binomtest

y = np.asarray(y_test)
pb, pi = np.asarray(baseline_test_prob), np.asarray(improved_test_prob)
cb, ci = ((pb >= .5) == y), ((pi >= .5) == y)
b, c = int((cb & ~ci).sum()), int((~cb & ci).sum())
print("only baseline correct b =", b, "| only improved correct c =", c,
      "| exact McNemar p =", binomtest(b, b + c, .5).pvalue)

rng = np.random.default_rng(42)
d_auc, d_pr = [], []
for _ in range(5000):
    i = rng.integers(0, len(y), len(y))          # resample the SAME images for both models
    if len(np.unique(y[i])) < 2: continue
    d_auc.append(roc_auc_score(y[i], pi[i]) - roc_auc_score(y[i], pb[i]))
    d_pr.append(average_precision_score(y[i], pi[i]) - average_precision_score(y[i], pb[i]))
for name, d in [("ROC-AUC", d_auc), ("PR-AUC", d_pr)]:
    lo, hi = np.percentile(d, [2.5, 97.5])
    p_two = 2 * min((np.array(d) <= 0).mean(), (np.array(d) >= 0).mean())
    print(f"{name} diff (improved - baseline): 95% CI [{lo:.4f}, {hi:.4f}], bootstrap p ~
{p_two:.4f}")
```

A.2 Choosing the threshold on validation data only

```
from sklearn.metrics import roc_curve
val_prob = improved.predict(val_ds, verbose=0).ravel()        # validation predictions
fpr, tpr, thr = roc_curve(y_val, val_prob)
k = np.argmax(tpr >= 0.95)          # first threshold reaching >= 95% malignant sensitivity
T = thr[k]; print("validation-chosen threshold:", T)
# Then evaluate ONCE on the test set: metric_row(y_test, improved_test_prob, "Improved @ T",
threshold=T)
```

Appendix B. How the McNemar bounds were derived

Let a = both correct, b = only baseline correct, c = only improved correct. From the confusion matrices: a + b = 439 and a + c = 322, so b − c = 117. Within the 78 benign images the baseline is right on 8 and the improved model on 77; within the 439 malignant images the baseline is right on 431 and the improved model on 245. The overlap of correct answers in each class can only lie between max(0, x + y − n) and min(x, y), which yields 237–245 both-correct malignant and 7–8 both-correct benign. This gives b between 186 and 195 and c between 69 and 78. Every admissible pair has the same b − c = 117, so $\chi^2 = 116^2 / (b + c)$ lies in [49.3, 52.8].